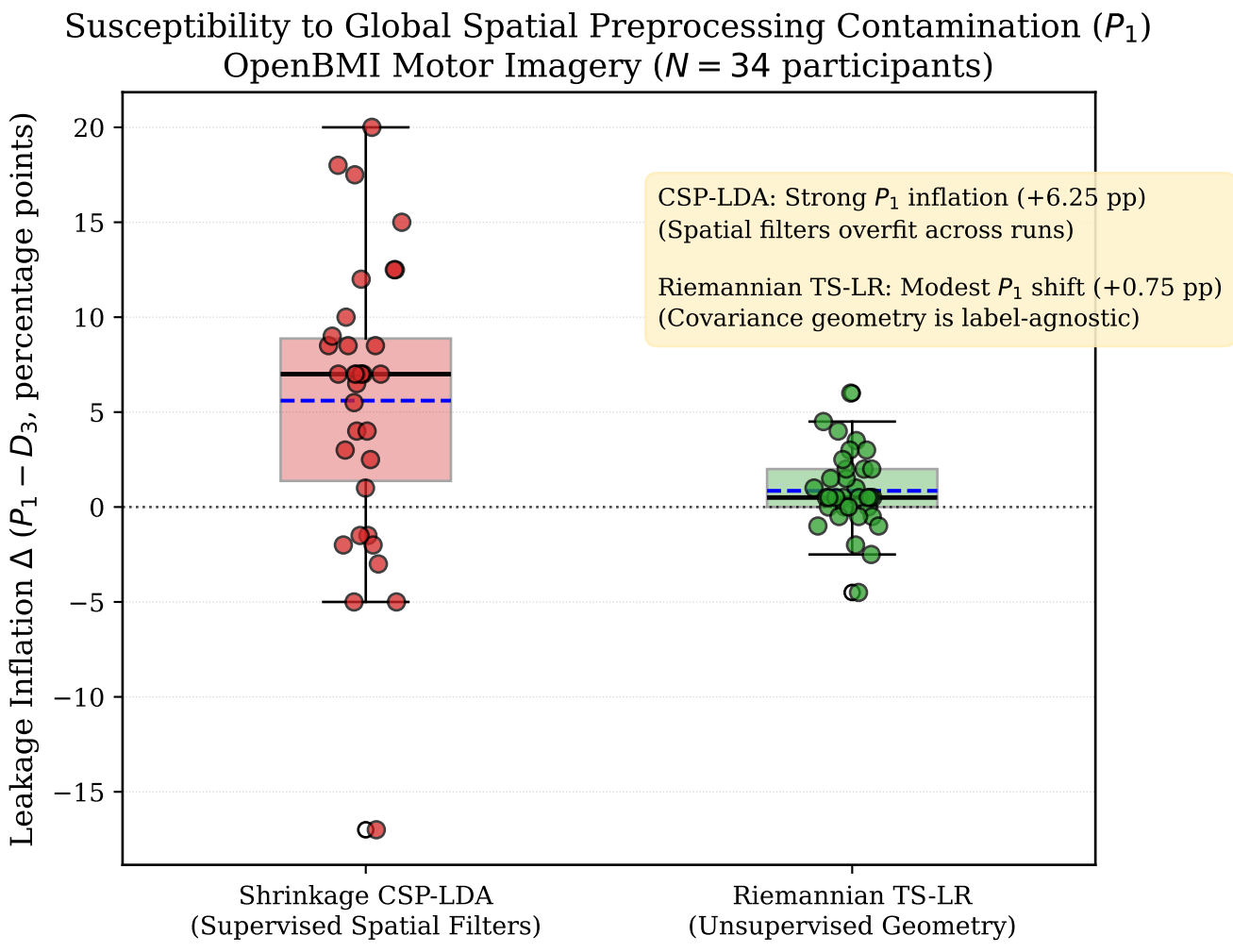
Susceptibility to Global Spatial Preprocessing Contamination ($P_1$)
OpenBMI Motor Imagery ($N = 34$ participants)
Leakage Inflation Δ ($P_1 - D_3$, percentage points)
20
15
10
5
0
−5
−10
−15
CSP-LDA: Strong $P_1$ inflation (+6.25 pp)
(Spatial filters overfit across runs)
Riemannian TS-LR: Modest $P_1$ shift (+0.75 pp)
(Covariance geometry is label-agnostic)
Shrinkage CSP-LDA
(Supervised Spatial Filters)
Riemannian TS-LR
(Unsupervised Geometry)